

J. SakethaNath

MACHINE LEARNING

Research

I work on generic machine learning formulations and algorithms that may have diverse applications. Recent topic of interest is constrained sampling using deep generative models. Some main themes I worked on earlier are: optimal transport, domain adaptation under label-shift, multiple kernel learning, robust optimization.

PhD Students

Associate Prof
@ IITH

Krishn Vishwas Kher (co-guiding with Prof. Vineeth NB)
Ongoing thesis; PMRF fellow

Piyushi Manupriya

Pursuing Postdoc at IISc

Thesis: "Improving Optimal Transport using Kernel Methods"
CSE@IITH nominee for ACM iKDD Doctoral Dissertation in Data Science (defense: 2025)

Associate Prof
@ IITB

Arun Iyer (co-guided with Prof. Sunita Sarawagi)
Principal Researcher @ Microsoft Research

Thesis on handling label-shift for domain adaptation
(defense: 2018)

Assistant Prof
@ IITB

Pratik Jawanpuria

Currently, Associate Professor, IIT Bombay

Thesis: "Learning kernels for multiple predictive tasks"
Received thesis awards from IIT Bombay and Xerox research
(defense: 2015)

Publications

Associate Professor (2015-present)

AAMAS'25

Piyushi Manupriya, Himanshu, SakethaNath Jagarlapudi and Ganesh Ghalme. Multi-agent Multi-armed Bandits with Minimum Reward Guarantee Fairness.
"A*" rated conference

ICLR'25

Piyushi Manupriya, J. Saketha Nath and Pratik Jawanpuria. MMD-Regularized Unbalanced Optimal Transport.
"A*" rated premier conference. (invited from TMLR'24)

ICML'24

Piyushi Manupriya, Pratik Kumar Jawanpuria, Karthik Gurumoorthy, Sakethanath Jagarlapudi and Bamdev Mishra.

	Submodular Framework for Structured-sparse Optimal Transport "A*" rated premier conference
AISTATS'24	Piyushi Manupriya, Rachit Keerti Das, Sayantan Biswas, and J. Saketha Nath. CONSISTENT OPTIMAL TRANSPORT WITH EMPIRICAL CONDITIONAL MEASURES. "A" rated premier conference
NPJ'24 Systems Biology and applications	Karthik Raman, Sowmya Manojna Narasimha, Tanisha Malpani, Omkar Mohite, and Saketha Nath. Understanding flux switching in metabolic networks through an analysis of synthetic lethals. 3.5 Impact factor
AISTATS'22	Piyushi Manupriya, SakethaNath Jagarlapudi, Tarun Ram Menta and Vineeth N. Balasubramanian. Improving Attribution Methods by Learning Submodular Functions. "A" rated premier conference
EACL'21	Soumya Chatterjee, Ayush Maheshwari, Ganesh Ramakrishnan and SakethaNath Jagarlapudi. Joint Learning of Hyperbolic Label Embeddings for Hierarchical Multi-label Classification. "A" rated conference
NeurIPS'20	J. Saketha Nath and Pratik Jawanpuria. Statistical Optimal Transport posed as Learning Kernel Embedding. "A*" rated premier conference
PRICAI'19	Prafull Prakash, Chaitanya Murti, J. Saketha Nath and Chiranjib Bhattacharyya. Optimizing DNN Architectures for High Speed Autonomous Navigation in GPS Denied Environments on Edge Devices "B" rated conference
NAACL-HLT'18	Ayush Maheshwari, Vishwajeet kumar, Ganesh Ramakrishnan and J. Saketha Nath. Entity Resolution and Location Disambiguation in Ancient Hindu Temples Domain using Web Data. "A" rated conference
KDD'16	Arun Iyer, Saketha Nath J and Sunita Sarawagi. Privacy-preserving Class Ratio Estimation. "A*" rated premier conference
Assistant Professor (2009-2015)	
JMLR'15	g. Generalized Hierarchical Kernel Learning. "A*" rated premier journal
ICML'14	Pratik J., Manik Varma and Saketha Nath J. On p-norm Path Following in Multiple Kernel Learning for Non-linear Feature Selection. "A*" rated premier conference
ICML'14	

	Arun Iyer, Saketha Nath J and Sunita Sarawagi. Maximum Mean Discrepancy for Class Ratio Estimation: Convergence Bounds and Kernel Selection. "A*" rated premier conference
Opt&Engg'13	J. Saketha Nath, A. Ben-Tal and C. Bhattacharyya. Robust formulations for clustering-based large-scale classification. Journal of Optimization & Engg. Impact factor: 1.7
ACL'13	Ankit Ramteke, Akshat Malu, Pushpak Bhattacharyya and Saketha Nath. Detecting Turnarounds in Sentiment Analysis: Thwarting. "A*" rated conference
ICML'12	Pratik J., and J. Saketha Nath. A Convex Feature Learning Formulation for Latent Task Structure Discovery. "A*" rated premier conference
ICML'11	Pratik J., J. Saketha Nath and Ganesh R. Efficient Rule Ensemble Learning using Hierarchical Kernels. "A*" rated premier conference
JMLR'11	J. Aflalo, A. Ben-Tal, C. Bhattacharyya, J. Saketha Nath and S. Raman. Variable Sparsity Kernel Learning "A*" rated premier journal
SDM'11	Pratik J., and J. Saketha Nath. Multi-task Multiple Kernel Learning. "A" rated conference
Math Prog'10 Series B	A. Ben-Tal, S. Bhadra, C. Bhattacharyya and J. Saketha Nath. Chance Constrained Uncertain Classification via Robust Optimization. Impact factor: 2.2
NeurIPS'09	J. Saketha Nath, G. Dinesh, S. Raman, C. Bhattacharyya, A. Ben-Tal, K. R. Ramakrishnan. On the Algorithmics and Applications of a Mixed-norm based Kernel Learning Formulation. "A*" rated premier conference
PAKDD'09 (best paper runner-up)	S. Bhadra, J. Saketha Nath, A. Ben-Tal and C. Bhattacharyya. Interval Data Classification under Partial Information: A Chance-Constraint Approach. "B" rated conference
ICML'07	R. Babaria, J. Saketha Nath, S. Krishnan, Sivaramakrishnan, C. Bhattacharyya and M. N. Murty. Focussed Crawling with Scalable Ordinal Regression Solvers. "A*" rated premier conference
SDM'07	J. Saketha Nath and C. Bhattacharyya. Maximum Margin Classifiers with Specified False Positive and False Negative Error Rates. "A" rated conference
KDD'06	J. Saketha Nath, C. Bhattacharyya and M. N. Murty. Clustering Based Large Margin Classification: A Scalable Approach using SOCP Formulation. "A*" rated premier conference

PatRec'06	J. Saketha Nath and S. K. Shevade. An efficient clustering scheme using support vector methods. "A*" rated journal
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Sponsored

Projects

Fujitsu 2022	Free grant of ~1 crore. Majorly utilized for compute infrastructure building for my lab.
Google 2021	Free grant of ~40lakhs as PhD award to Piyushi. Mainly utilized for supporting PhD fellowship.
Microsoft 2012	Free grant of ~12 lakhs for supporting research in my lab.

Teaching

Involved in floating new courses: CS5590 (core), CS6090, CS5660, CS5580, CS5560, CS5470.
Average teaching score = 4.19/5

2025	Aug-Nov	CS5590/CS3390	Foundations of Machine Learning
2025	Jan-Apr	CS6090	Deep Generative Models
2024	Aug-Nov	ID	Introduction to Programming
2024	Aug-Nov	CS6660	Mathematical Foundations of Datascience
2024	Jan-Apr	CS5660	Convex Optimization 2: Algorithms
2023	Aug-Nov	CS5580	Convex Optimization 1: Theory
2023	Jan-Apr	CS5470	Theory of Learning and Kernels
		CS5590/CS3390/AI2000/AI	
2022	Aug-Nov	5000	Foundations of Machine Learning
2022	Jan-Apr	CS5560	Probabilistic Models for Machine Learning
2021	Aug-Nov	CS5580	Convex Optimization 1: Theory
2021	Jan-Apr	CS5560	Probabilistic Models for Machine Learning
2020	Aug-Nov	CS5470	Theory of Learning and Kernels
2020	Jan-Apr	CS5660	Convex Optimization 2: Algorithms
2019	Aug-Nov	CS5590	Foundations of Machine Learning
2019	Jan-Apr	cs5580	Convex Optimization: Theory
2018	Aug-Nov	CS5560	Probabilistic Models for Machine Learning
2018	Jan-Apr	cs1340	discrete structures
2017	Aug-Nov	CS5470	Theory of Learning and Kernels

Professional service

Area chair and meta-reviewer for premier conferences like NeurIPS, ICML etc.

References

Prof. M. Vidyasagar	Fellow of The Royal Society Distinguished Professor @ IITH and formerly SERB National Science Chair [m.vidyasagar@ee.iith.ac.in] [+91 9848050451]
Prof. Bharat Sriperumbudur	Professor, Departments of Statistics and Mathematics, Pennsylvania State University. NSF CAREER Awardee, Associate Editor: JMLR, Annals of Statistics, and Bernoulli. [bks18@psu.edu] [(814) 867-5948]
Prof. Sunita Sarawagi	Professor, CSE, IIT Bombay ACM Fellow, Infosys Prize 2019 recipient IITKGP Distinguished Alumnus [sunita@iitb.ac.in] [+91 9969345905]
Prof. Chiranjib Bhattacharyya	Professor, CSA, IISc Fellow of Indian Academy of Engg. [chiranjib.bhattacharyya@gmail.com] [chiru@iisc.ac.in] [+91 9980017564]